

Generation Interconnection System Impact Study Report

For Queue Project Q18-1049

Stent 138 kV
February 19, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1049. The Interconnection Customer proposes a 273.4 MW wind facility interconnecting at the Stent 138 kV substation in the Pennsylvania area. The request was received on August 29, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 350 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,304,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: STENT - BARBADOES 138.0 kV
Contingency: STARKEY ROAD - PICO 138.0 kV
Loading: 139.4% (Rating: 212 MVA)
DFAX: 57.8%
Pre-project loading: 64.8%
2. Facility: FRUITLAND SUBSTATION - ST. JOHNS 138.0 kV Line
Contingency: MARCUS HOOK ENERGY LP - TOMBSTONE SUBSTATION 500.0 kV
Loading: 118.2% (Rating: 233 MVA)
DFAX: 10.5%
Pre-project loading: 105.9%
3. Facility: ST. JOHNS - HATHAWAY STREET 138.0 kV
Contingency: MARCUS HOOK ENERGY LP - TOMBSTONE 500.0 kV
Loading: 113.7% (Rating: 273 MVA)
DFAX: 10.5%
Pre-project loading: 103.2%
4. Facility: PICO - STENT 138.0 kV
Contingency: BARBADOES - STENT 138.0 kV
Loading: 108.6% (Rating: 224 MVA)
DFAX: 100.0%
Pre-project loading: 13.5%
5. Facility: DORCHESTER - HATHAWAY ST 138.0 kV
Contingency: TOMBSTONE - MARCUS HOOK ENERGY LP 500.0 kV
Loading: 108.2% (Rating: 349 MVA)
DFAX: 10.5%
Pre-project loading: 100.0%
6. Facility: FRUITLAND - EAST BENCH 138.0 kV L
Contingency: MARCUS HOOK ENERGY LP - TOMBSTONE 500.0 kV L
Loading: 107.5% (Rating: 229 MVA)

DFAX: 10.5%

Pre-project loading: 95.0%

7. Facility: DORCHESTER SUBSTATION - CECIL FIELD 138.0 kV

Contingency: TOMBSTONE - MARCUS HOOK ENERGY LP 500.0 kV Line

Loading: 107.0% (Rating: 370 MVA)

DFAX: 10.5%

Pre-project loading: 99.3%

8. Facility: CECIL FIELD - PLYMOUTH MEETING 138.0 kV

Contingency: TOMBSTONE - MARCUS HOOK ENERGY LP 500.0 kV

Loading: 106.3% (Rating: 384 MVA)

DFAX: 10.5%

Pre-project loading: 98.9%

9. Facility: PLYMOUTH MEETING - BARBADOES 138.0 kV Line

Contingency: TOMBSTONE - MARCUS HOOK ENERGY LP 500.0 kV L

Loading: 105.5% (Rating: 402 MVA)

DFAX: 10.5%

Pre-project loading: 98.4%

10. Facility: E BENCH - STILLWATER 138.0 kV

Contingency: TOMBSTONE - MARCUS HOOK ENERGY LP 500.0 kV

Loading: 100.1% (Rating: 220 MVA)

DFAX: 10.5%

Pre-project loading: 87.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BARBADOES - STENT 138.0 kV L	Rebuild line with higher-capacity conductor	\$6,040,000	74% (\$4,497,000)
ST. JOHNS - FRUITLAND 138.0 kV	Rebuild line with higher-capacity conductor	\$3,077,000	63% (\$1,929,000)
ST. JOHNS - HATHAWAY ST 138.0 kV	Rebuild line with higher-capacity conductor	\$1,544,000	63% (\$967,000)
PICO - STENT 138.0 kV	Reconductor line	\$387,000	100% (\$387,000)
DORCHESTER - HATHAWAY ST 138.0 kV	Reconductor line	\$512,000	63% (\$321,000)
FRUITLAND - EAST BENCH 138.0 kV L	Reconductor line	\$593,000	63% (\$371,000)
DORCHESTER - CECIL FIELD 138.0 kV	Reconductor line	\$663,000	63% (\$415,000)
PLYMOUTH MEETING - CECIL FIELD SUBSTATION 138.0 kV	Reconductor line	\$634,000	63% (\$397,000)
PLYMOUTH MEETING - BARBADOES 138.0 kV Line	Reconductor line	\$1,303,000	63% (\$817,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
STILLWATER - EAST BENCH 138.0 kV	Reconductor line	\$325,000	63% (\$203,000)
	Total allocated to project		\$10,304,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.